# Interface Keepalive Configuration

## 1. Introduction

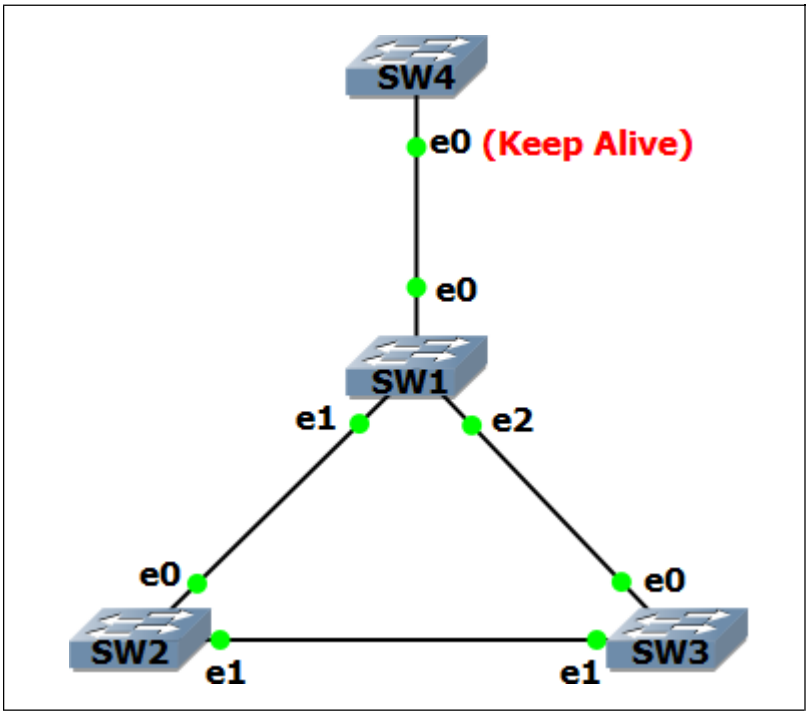
*keepalive* is a message sent by one device to another to check that the link between the two is operating.

## 2. Description

A ***keepalive*** signal is often sent at predefined intervals and plays an important role in the network. After a signal is sent, if no reply is received the link is assumed to be down. The ***keepalive*** command specifies the interval (in seconds) that the Switch waits before sending a message on the interface to test the link and determine whether it is up or down. The Switch produces an Ethernet packet (Which **source** and **destination** MAC-address is same) and a special Ethernet type code of 0x9000. The Switch sends this packet through the configured interface and then immediately receives this packet back again.

## 3. Topology

SW1, SW2 and SW3 creates a loop, i.e it is a ring network. If there is any loop in this network SW4 will be affected. To protect this "***keepalive***" command is used. In this scenario if this command is implemented SW4 will not be affected rather only port e0 of SW4 will be shuut down.



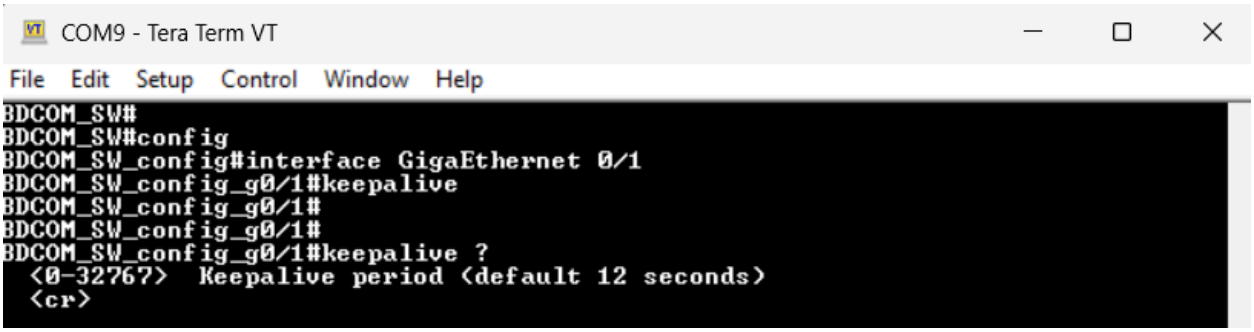
## 4. Configure Keepalive Settings

BDCOM_SW#config

BDCOM_SW_config#interface GigaEthernet 0/1

BDCOM_SW_config_g0/1#keepalive

[**Note:** By default, Keepalive period is 12 seconds. We may change the keepalive time accordingly with the command **keepalive <interval>**]




## 5. Verify Keepalive Configuration

show running-config <interface-name>

THE END